

Designation: E 2567-24

Standard Specification for Lithium-Ion Battery Cells for Electric Vehicle Applications

This standard was issued on March 5, 2024; the version of record is continuously maintained.

1. Scope

1.1 This specification covers high-energy density lithium-ion cylindrical cells for electric vehicle battery packs

1.2 The lithium-ion battery cells covered by this specification are intended for use in powering battery electric vehicles (BEVs) and plug-in hybrid electric vehicles (PHEVs) requiring high energy storage capacity and power delivery.

1.3 The values stated in SI units are to be regarded as standard. No other units of measurement are included in this standard.

1.4 This standard does not purport to address all of the safety concerns, if any, associated with its use. It is the responsibility of the user of this standard to establish appropriate safety, health, and environmental practices and determine the applicability of regulatory limitations prior to use.

2. Referenced Documents

2.1 ASTM Standards:

E 2950 Standard Guide for Chemical Analysis of Lithium-Ion Battery Materials

D 7032 Test Method for Determination of Fluorine in Aromatic Compounds

IEC 62660 Secondary lithium-ion cells for the propulsion of electric road vehicles

UN 38.3 Recommendations on the Transport of Dangerous Goods, Manual of Tests and Criteria

3. Terminology

3.1 Definitions:

3.1.1 nominal capacity—the charge capacity of a cell measured under standard discharge conditions

3.1.2 C-rate—a measure of charge or discharge current relative to cell capacity

3.1.3 state of charge—the percentage of available capacity remaining in a cell

4. Classification

4.1 lithium-ion battery cells shall be classified according to cell format, capacity range, and cathode chemistry.

4.1.1 **Format A**—Cylindrical 18650, capacity 2.5-3.5 Ah, NMC cathode

4.1.2 **Format B**—Cylindrical 21700, capacity 4.0-5.0 Ah, NMC cathode

4.1.3 **Format C**—Cylindrical 46800, capacity 20-25 Ah, NCA cathode

5. Ordering Information

5.1 Orders for material under this specification shall include the following information:

5.1.1 ASTM designation and year of issue

5.1.2 Cell format (A, B, or C)

5.1.3 Minimum capacity (Ah)

5.1.4 Quantity (number of cells)

5.1.5 Voltage and temperature specifications

5.1.6 Quality grade (automotive or industrial)

6. Materials and Manufacture

6.1 Cathode active material shall be lithium nickel manganese cobalt oxide (NMC) or lithium nickel cobalt aluminum oxide (NCA).

6.2 Anode shall be graphite-based with optional silicon content not exceeding 5%.

6.3 Electrolyte shall be lithium hexafluorophosphate (LiPF₆) in organic carbonate solvents.

6.4 Separator shall be tri-layer polyolefin with ceramic coating.

7. Chemical Composition

7.1 The material shall conform to the chemical composition requirements specified in Table 1.

Element	Composition, %
Cathode (NMC 811)	LiNi _{0.8} Mn _{0.1} Co _{0.1} O ₂
Anode	Graphite (95-100%) + Si (0-5%)
Electrolyte	1M LiPF ₆ in EC:EMC (3:7)
Moisture content	< 20 ppm

8. Mechanical and Physical Requirements

Property	Requirement	Test Method
Nominal capacity	≥ Rated capacity	IEC 62660-1
Energy density	≥ 250 Wh/kg	Calculated
Cycle life (80% retention)	≥ 1000 cycles	IEC 62660-1
DC resistance	≤ 20 mΩ	IEC 62660-1

9. Dimensions and Permissible Variations

9.1 Cell dimensions shall conform to standard battery formats with ±0.5 mm tolerance.

9.2 18650: Diameter 18.4 ±0.2 mm, Length 65.2 ±0.5 mm

9.3 21700: Diameter 21.0 ± 0.2 mm, Length 70.5 ± 0.5 mm

9.4 Weight variation shall not exceed $\pm 3\%$ within a production lot.

10. Workmanship, Finish, and Appearance

10.1 Cells shall be free from physical damage, electrolyte leakage, and terminal defects. Insulation sleeve shall be properly applied and free from tears.

11. Sampling

11.1 Sample size shall follow ANSI/ASQC Z1.4 acceptable quality level of 0.25%. Each lot shall undergo safety and performance testing.

12. Test Methods

12.1 **Capacity Testing**—Fully charge cells to 4.2V, then discharge at 0.5C to 2.5V while measuring capacity.

12.2 **Cycle Life Testing**—Cycle cells between 100% and 0% SOC at 1C rate until capacity drops to 80% of initial.

12.3 **Safety Testing**—Conduct overcharge, short circuit, crush, and thermal abuse tests per UN 38.3 and IEC 62660-2.

13. Inspection

13.1 Inspection of the material shall be made as agreed upon between the purchaser and the seller as part of the purchase contract.

14. Rejection and Rehearing

14.1 Material that fails to conform to the requirements of this specification may be rejected. Rejection should be reported to the producer or supplier promptly and in writing.

15. Certification

15.1 A manufacturer's or supplier's certification shall be furnished to the purchaser that the material was manufactured, sampled, tested, and inspected in accordance with this specification and has met the requirements.

16. Product Marking

16.1 Each cell shall be marked with manufacturer name, model number, rated capacity, rated voltage, and date code. Polarity shall be clearly indicated.

17. Packaging and Package Marking

17.1 Cells shall be packaged at 30-50% state of charge in anti-static trays with protective caps. Packages shall comply with UN 38.3 shipping requirements.

18. Keywords

18.1 lithium-ion; battery; electric vehicle; energy storage; rechargeable; NMC; automotive